

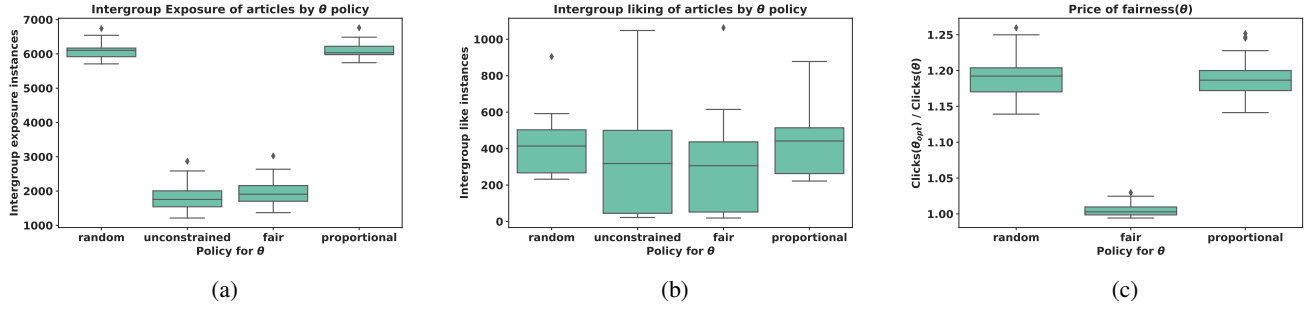


Figure 3: Intergroup exposure (a) and liking (b), as well as the price of fairness (c), using model parameters from ?.

As the Fairness Doctrine was introduced to ensure opinion diversity, a modern version of this policy could be similarly introduced in online platforms (?). Although both the Fairness Doctrine and our model expose users of different groups to a diverse set of news articles at the same time, our model might expose each individual to only one article. Such platform interventions have the great potential to ensure diversity of viewpoints; however, the design of such policies entails the careful examination of any ethical concerns. A question that naturally arises is whether it is ethical for the platform to algorithmically control and potentially randomize the content that a user sees and, ultimately, who—if anyone—has the responsibility to ensure fair exposure in online spaces. This question has been under close scrutiny in interpreting Section 230 of the Communications Decency Act in the United States (?). For example, given that news sharing and discussions in social media can determine important political outcomes and thus the passive or more restrictive role that the platform chooses to undertake matters (see, e.g., ??), it is unclear how a fair representation of content should be defined. Thus, we acknowledge that balancing exposure to different ideologies of content might not actually be fair in a given context. For instance, ? suggest that showing people opposing viewpoints makes them more polarized, whereas ? show that echo chambers do not necessarily increase polarization. Furthermore, considering the amount of disinformation and the technical challenges in identifying problematic content (e.g., fake news, hate speech) in platforms, the fair exposure constraints should not be applied to all content. Implementing fair exposure can thus become particularly challenging, and more interdisciplinary research is needed to understand where to draw the boundary. Our work is an initial step towards this broader goal.

Finally, our framework highlights how the introduction of fairness constraints can only partially mitigate group-homogeneous targeting and points to problematic outcomes, as sometimes only one group incurs the “price of fairness” while the other pays the “cost of user engagement.” It also gives rise to a series of emerging, challenging directions for future research related to platforms and algorithmic fairness. These include the study of fair exposure notions, the design of dynamic interventions and more sophisticated targeting, ad pricing and revenue maximization under fair exposure

constraints, and their implications on the competition among different platforms.

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